1 equations

ALL IN THIS 1-2:

$\frac{A_2}{B_2} = 8\pi\bar{h}\frac{\mu^3}{\lambda^3}$;

$g_1 B_{12} = g_2 B_{21}$

$-dn_2(\nu) = A_{21} n_2 f(\nu) d\nu$;

$-dn_2 = B_{21} n_2 f(\nu) \rho_\nu d\nu$

$f(\nu) = \frac{\frac{\Delta\nu}{2\pi}}{(\nu-\nu_0)^2+(\frac{\Delta\nu}{2})^2}$;

$\Delta\nu = \frac{1}{2\pi\tau}$;

GAP$= \sqrt{1+\frac{I}{I_s}}\Delta\nu$;

Stable $\boldsymbol{Condition}$:$0 < (1-\frac{L}{R_1})(1-\frac{L}{R_2}) < 1$;

$\Delta n = \frac{[(\nu-\nu_0)^2+(\frac{\Delta\nu}{2})^2]\Delta n_0}{(\nu-\nu_0)^2+(1+\frac{I}{I_s})(\frac{\Delta\nu}{2})^2} = \frac{\Delta n_0}{1+\frac{I}{I_s}\frac{f(\nu)}{f(\nu_0)}}$;

$\Delta n = n_2 - \frac{g_2}{g_1} n_1$;

$G = \Delta n B_{21} f(\nu) h\nu \frac{mu}{c}$;

$\Delta n = R_2\tau_2 - (R_1+R_2)\tau_1$;

$G = \frac{[(\nu-\nu_0)^2+(\frac{\Delta\nu}{2})^2]G^0}{(\nu-\nu_0)^2+(1+\frac{I}{I_s})(\frac{\Delta\nu}{2})^2} = \frac{G^0}{1+\frac{I}{I_s}\frac{f(\nu)}{f(\nu_0)}}$;

$G^0(\nu) = \Delta n_0 B_{21}\frac{\mu}{c} f(\nu) h\nu_0$

$\Delta n(\nu) = \frac{[(\nu-\nu_0)^2+(\frac{\Delta\nu}{2})^2]\Delta n_0 f_D(\nu)}{(\nu-\nu_0)^2+(1+\frac{I}{I_s})(\frac{\Delta\nu}{2})^2} = \frac{\Delta n_0 f_D(\nu)}{1+\frac{I}{I_s}\frac{f(\nu)}{f(\nu_0)}}$;

wide of hole$\nu \pm \sqrt{1+\frac{I}{I_s}}\frac{\Delta\nu}{2}$

$G_D(\nu_1) = \frac{G_d^0(\nu_1)}{\sqrt{1+\frac{I}{I_s}}}$

$a_{all} = a_{in} - \frac{1}{2L} ln r_1 r_2 = -\frac{1}{2L} ln(1-(a_1+t_1))$

$\boldsymbol{Condition}$:$2\delta\Phi = 2q\pi, q = 1, 2, 3...$

$\Delta\nu_q = \frac{c}{2\mu L}$

$2L = q\lambda$

$\boldsymbol{s}quare\Delta\Phi_{mn} = (m+n+1)\frac{\pi}{2}$

$\nu_{nmq} = \frac{c}{2\mu L}(q+\frac{1}{2}(m+n+1))$

$\frac{1}{2}\Delta\nu_q = \Delta\nu_m = \Delta\nu_n$

$\boldsymbol{r}ound\Delta\Phi_{mn} = (m+2n+1)\frac{\pi}{2}$

$\nu_{nmq} = \frac{c}{2\mu L}(q+\frac{1}{2}(m+2n+1))$

$\frac{1}{2}\Delta\nu_q = \Delta\nu_m = \frac{1}{2}\Delta\nu_n$

$\boldsymbol{G}auss$:

$\omega(z) = \omega_0\sqrt{1+(\frac{f}{z})^2}$

$\omega_0 = \sqrt{\frac{\lambda f}{\pi}}$

$R(z) = z[1+(\frac{f}{z})^2]$

$L = |z_1| + |z_2|$

$R_1 = R_2 = 2f$(S.C chamber) not suitable for normal shpere.

$f = \frac{\pi\omega_0^2}{\lambda}$

$2\theta = \frac{2\lambda}{\pi\omega_0}$

$q = z + if$
$2\theta_{mn} = \sqrt{2m+1}\sqrt{2n+1}2\theta_0$
${V_{00}}^0 = \frac{L^2\lambda}{2}$
${V_{mn}}^0 = \sqrt{2m+1}\sqrt{2n+1}{V_{00}}^0$
HOMOS
$G = \frac{G^0}{1+2\frac{I}{I_s}} = G_{Thres}$
$a_{all} = \frac{a_1+t_1}{2L}$
$I = \frac{I_s}{2}t_1(\frac{2LG^0}{a_1+t_1} - 1)$
$P = IS$
non-HOMOS
non-center franquency :$G(\nu) = \frac{G_D^0(\nu)}{sqrt1+\frac{I}{I_s}} = G_{Thres}$
center franquency :$G(\nu) = \frac{G_D^0(\nu)}{sqrt1+2\frac{I}{I_s}} = G_{Thres}$

2 homeworks

all:

chapter1: 2,3,4,11,12,13 P28
chapter2: 1,2,3,7,8,10,11,12,13 P48
chapter3: 1,2,4,6,7,9,10 P91
chapter4: 1,2,3,4,5,8,9 P119

3 keys in chapter 1

CHAPTER1 Formulas used in homework.

$$n \sim g e^{-\frac{E}{kT}} \tag{1}$$

$$\frac{q_s}{q_{normal}} = \frac{c^3}{8\pi h\nu^3} = \frac{1}{e^{\frac{h\nu}{kT}} - 1} \tag{2}$$

$$I = I_0 e^{-Gz} \tag{3}$$

$$\nu = \nu_0(1 + \frac{v}{c}) \tag{4}$$

$$f(\nu) = \frac{\frac{\Delta\nu}{2\pi}}{(\nu - \nu_0)^2 + (\frac{\Delta\nu}{2})^2} \quad \Delta\nu = \frac{1}{2\pi\tau} \tag{5}$$

$$\rho(\nu) = \frac{\frac{8\pi h v^3}{c^3}}{e^{\frac{h\nu}{kT}} - 1} \tag{6}$$

$$\tag{7}$$

SOME **PROPOSITION NEEDED TO BE PROVED IN CHAPTER 1:**

I:PROVE:

$$A = \frac{1}{\tau} \tag{8}$$

prove:

$$n_2(t) = n_{20} e^{At}$$
$$\text{consider n}: \; n = n_0 e^{\frac{t}{\tau}}$$
$$\text{so}: \; A_{21} = \frac{1}{\tau}$$

II:PROVE:

$$A_{21} = B_{21} \frac{8\pi h\nu^3}{c^3} \tag{9}$$

$$B_{21} = B_{12} \frac{g_1}{g_2} \tag{10}$$

prove:

$$A_{21} n_2 dt + B_{21} n_2 \rho_\nu dt = B_{12} n_1 \rho_\nu dt$$
$$(A_{21} + B_{21}\rho_\nu)\frac{n_2}{n_1} = B_{12}\rho_\nu$$

consider plank ρ_ν :

$$n \sim g e^{\frac{-E}{kT}}$$

$$\frac{n_2}{n_1} = \frac{g_2}{g_1} e^{-\frac{h\nu}{kT}}$$

$$A_{21} = (\frac{g_1}{g_2} e^{\frac{h\nu}{kT}} B_{12} - B_{21})\rho_\nu$$

$$\rho_\nu = \frac{A_{21}}{\frac{g_1}{g_2} e^{\frac{h\nu}{kT}} B_{12} - B_{21}}$$

$$\rho_\nu = \frac{A_{21}\frac{g_2}{g_1 B_{12}}}{e^{\frac{h\nu}{kT}} - \frac{g_2}{g_1 B_{12}} B_{21}}$$

compare with : $\rho_\nu = \frac{\frac{8\pi h\nu^3}{c^3}}{e^{\frac{h\nu}{kT}} - 1}$

so we have:

$$\frac{8\pi h\nu^3}{c^3} = A_{21}\frac{g_2}{g_1 B_{12}} \tag{11}$$

$$1 = \frac{B_{21} g_2}{B_{12} g_1} \tag{12}$$

4 keys in chapter 2

SOME KEYS IN HOMEWORK:

$$0 < g_1 g_2 < 1; g_1 = 1 - \frac{L}{R_1}, g_2 = 1 - \frac{L}{R_2}$$

$$G = \Delta h\nu \frac{\mu}{c} f(\nu) B_{21}$$

$$\Delta n_0 = R_2\tau_2 - (R_1 + R_2)\tau_1$$

$$\Delta n = \frac{\Delta n_0}{1 + \frac{I f(\nu)}{I_s f(\nu_0)}}$$

$$G = \frac{G_0(\nu)}{1 + \frac{I f(\nu)}{I_s f(\nu_0)}} = \frac{(\frac{\Delta\nu}{2})^2 G_0(\nu_0)}{(\nu - \nu_0)^2 + (1 + \frac{I}{I_s})(\frac{\Delta\nu}{2})^2} \quad \Delta\nu' = \sqrt{1 + \frac{I}{I_s}}\Delta\nu$$

$$G_D = \frac{G_0(\nu) f_D(\nu)}{1 + \frac{I f(\nu)}{I_s f(\nu_0)}}$$

$$G_D(\nu_1) = \frac{G_D^0(\nu_1)}{(1 + \frac{I}{I_s})^{\frac{1}{2}}}$$

$$a_{whole} = a_{iner} - \frac{1}{2L}ln(r_1r_2)$$
$$T_L = \begin{bmatrix} 1 & L \\ 0 & 1 \end{bmatrix}; T_f = \begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix}; T_{\frac{n_1}{n_2}} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{n_1}{n_2} \end{bmatrix}$$

5 keys in chapter 3

$$R = z + \frac{f}{z}$$
$$\omega = \omega_0\sqrt{1 + (\frac{z}{f})^2}$$
$$\theta_0\omega_0 = \frac{\lambda}{\pi}$$
$$\omega_0 = \sqrt{\frac{\lambda f}{\pi}}$$
$$q = z + if$$
$$M^2 = \frac{\omega\theta}{\omega_0\theta_0}$$
$$P_o = \frac{1}{2}t_1 I_s A(\frac{2LG_0}{a_1 + t_1} - 1)$$
$$P_e(\nu_1) = t_1 I_s A[(\frac{2LG_D^0}{a_1 + t_1})^2 - 1]; P_e(\nu_0) = \frac{1}{2}I_s A t_1[(\frac{2LG_D^0(\nu_0)}{a_1 + t_1}) - 1]$$
$$\text{best t :}\frac{dP}{dt_1} = 0$$
$$\text{square:}\phi_{mn} = \frac{c}{2\mu L}(q + \frac{1}{2}(m + n + 1))$$
$$\text{round:}\phi_{mn} = \frac{c}{2\mu L}(q + (m + 2n + 1))$$
$$\text{LG:like square}\omega_{mn} = \sqrt{m + n + 1}\omega_{00}$$
$$\text{LG:}\theta_{00} = \sqrt{m + n + 1}\theta_{00}$$
$$\text{HG:Hermit Gauss :}\omega_m = \sqrt{2m + 1}\omega_{00}; \omega_n = \sqrt{2n + 1}\omega_{00}$$
$$\text{HG:Hermit Gauss :}\theta_m = \sqrt{2m + 1}\theta_{00}; \theta_n = \sqrt{2n + 1}\theta_{00}$$

6 keys in chapter 4

$$\tau = \frac{T}{2N+1}$$
$$P_{locked} = NP_{aver} = N^2 P_{single}$$